M. Ikram Ullah

Data Scientist · Experimentation, Causal Inference & Machine Learning

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SUMMARY

Data scientist who turns experiments and data into decisions with honest uncertainty. Recent work: a full readout of a 64,000-customer randomized e-mail experiment (validity checks, Holm-corrected effects, CUPED, power and peeking analysis, uplift models and an IPW-valued targeting policy), a text-to-SQL analyst agent benchmarked on BIRD, and a retrieval benchmark over 84 company filings. For clients I have built churn, fraud, stroke-risk, forecasting and recommendation models end to end. Strong in Python, SQL and statistics; experienced with dashboards for non-technical stakeholders; pursuing an MS in Data Science.

TECHNICAL SKILLS

Statistics & experimentation: A/B testing, sample-ratio mismatch, power analysis and MDE, CUPED and regression adjustment, bootstrap, multiple-testing correction (Holm), sequential-testing pitfalls, causal inference

Machine learning: uplift modeling (S/T/X-learners, transformed outcome, Qini/AUUC), XGBoost, LightGBM, scikit-learn, time series (ARIMA, SARIMA, Prophet, LSTM), recommender systems (SVD, KNN), topic modelling (LDA, BERTopic), class-imbalance handling, SHAP/LIME, model monitoring

Data & SQL: advanced SQL (window functions, CTEs, query optimisation), DuckDB, SQLite, PostgreSQL, MySQL, MongoDB, pandas, NumPy, statsmodels, data pipelines

Visualisation & BI: Power BI (DAX), Tableau, Plotly, matplotlib, Streamlit dashboards

AI & engineering: LLM applications (text-to-SQL, RAG), LangGraph, Gemini/OpenAI APIs, FastAPI, Docker, Git, pytest

EXPERIENCE

Data Scientist / ML Engineer, Syslab Technologies *(Islamabad, Pakistan)* Oct 2024 – Nov 2025

- Improved diagnostic response accuracy by 23% through systematic A/B testing of model configurations, prompt strategies and RAG retrieval settings (chunk size, retrieval depth).
- Built an automated model-monitoring pipeline (accuracy, precision, recall, F1, latency) with real-time Power BI dashboards for early detection of model drift.
- Investigated model failures through error logs, edge cases and interaction data; targeted retraining and data-quality fixes reduced the error rate by 18%.
- Processed 50K+ patient interaction records with Python and SQL; validated outputs with product managers, QA and healthcare providers against clinical standards in an Agile team.

Freelance Data Scientist, Self-employed (direct clients) *(Remote)* Feb 2023 – Present

- Delivered 15+ end-to-end AI/ML projects for healthcare, e-commerce and fintech clients, from scoping and data collection to deployment and handoff.
- Built churn, demand-forecasting and fraud-detection pipelines (scikit-learn, XGBoost, LightGBM) with MLflow experiment tracking; reported results through Tableau and Power BI dashboards.

PROJECTS

Data Analyst Agent (text-to-SQL) *(github.com/ikram4748/data-analyst-agent)*
LangGraph · Gemini · sqlglot · DuckDB · fastembed · Streamlit

- Built a self-correcting text-to-SQL agent: local-embedding schema linking, a two-layer read-only SQL guard, a literal-value check that catches silent zero-result errors, and a bounded fix loop driven by database feedback.
- Raised execution accuracy on 150 held-out BIRD questions from 60.7% (one-shot baseline, same model) to 66.7% (+6.0 points, 95% CI +1.3 to +11.3) at 1.09 LLM calls per question; tuned on a separate dev split after error analysis of 48 failures.
- Found and fixed a measurement bug in which rate-limit fallback silently mixed three models across benchmark variants.

Financial RAG Lab *(github.com/ikram4748/financial-rag-lab)*
PyMuPDF · BM25 · bge-small · cross-encoders · Gemini

- Built a RAG benchmark on FinanceBench (150 analyst questions, 84 filings, 12,013 pages) comparing 3 chunking strategies, 3 metadata scopes and 5 retrievers (BM25, dense, hybrid RRF, two cross-encoder rerankers) plus HyDE, scored on evidence-page recall with a dev/test split.

- On 100 held-out questions, dense retrieval with company/year filtering beat BM25, hybrid and cross-encoder reranking; HyDE raised evidence recall@5 from 44% to 60% and answer accuracy from 47% to 61% (+14 points, 95% CI +5 to +23), vs 24% for naive RAG and 84% with the evidence pages given, which showed retrieval was the bottleneck. Answers graded by a separate judge model validated on gold and corrupted answers.

Qanoon Assist: legal-information RAG for Pakistani law (private) *(code available on request)*
FastAPI · React · PostgreSQL · Qdrant · BGE-M3 · Gemini

- Scraped and parsed 6,664 federal and provincial laws (117,365 sections, 147,701 chunks) from five government portals; answers in English, Urdu and Roman Urdu with section-level citations.
- Hybrid BM25 + BGE-M3 dense retrieval with bge-reranker-v2-m3 and citation verification; hit@5 1.00, faithfulness 0.944 and 4.8 s average latency on a 36-question evaluation; 124 automated API and UI tests.

E-mail Experiment & Uplift Modeling *(github.com/ikram4748/uplift-ab-testing)*
Python · statsmodels · LightGBM · Plotly · Streamlit

- Analysed a 64,000-customer randomized e-mail experiment end to end: sample-ratio and balance checks, Holm-corrected effects (men's e-mail +7.7 points visits, +$0.77 spend per customer, 95% CI $0.49–$1.05), CUPED and regression adjustment, power analysis, and a peeking simulation (10 looks → 18% false positives).
- Built S/T/X-learner and transformed-outcome uplift models with Qini evaluation over 5 splits and valued targeting policies by inverse-propensity weighting; showed that e-mailing everyone beats model targeting below about $0.23 per e-mail.

CLIENT PROJECTS (CODE PRIVATE)

- **Autonomous Data Analyst** *(Gemini · pandas · sandboxed exec)*: Answers plain-English questions about any spreadsheet by having the LLM write pandas code that runs in a restricted sandbox; shows the computed result and the code.
- **Stroke-Risk Prediction** *(scikit-learn · XGBoost · imbalanced-learn · SHAP)*: Random Forest vs XGBoost on patient data with a 5% positive class; SMOTE on the training split only (no leakage), decision threshold tuned on the precision-recall curve, SHAP explanations per patient.
- **ChurnGuard (customer churn)** *(scikit-learn · XGBoost · SMOTE · Streamlit)*: Churn model on 7,043 telecom customers comparing four models (Random Forest best, 0.847 AUC-ROC), with an interactive form that returns churn risk and retention actions.
- **FraudShield (fraud detection)** *(XGBoost · LightGBM · SMOTE · SHAP · LIME)*: Fraud detection on 50K transactions with a 1.7% fraud rate; engineered velocity and risk features, five models with threshold optimisation and an ensemble, SHAP and LIME explanations for every prediction.
- **TimeVision (energy forecasting)** *(statsmodels · Prophet · LSTM)*: ARIMA, SARIMA, Prophet and LSTM on 3+ years of daily energy demand with stationarity tests and walk-forward validation; seasonal models clearly beat plain ARIMA.
- **RecEngine (recommendations)** *(scikit-learn · SVD · KNN)*: Content-based, user/item KNN, SVD and hybrid recommenders on 80K ratings at 87% sparsity, ranked by Precision@K, Recall@K and NDCG; SVD was best and the hybrid handles cold-start users.
- **TextMiner (summarization and topics)** *(T5 · TextRank · LDA · BERTopic)*: Extractive (TextRank, LSA) vs abstractive (T5) summarization scored with ROUGE, and LDA vs BERTopic topic models checked against true categories (ARI, NMI) on 300+ news articles.

RESEARCH

UrduComicLab: complete-story Urdu story-to-comic generation (MS thesis, in progress)

- Open-weight pipeline turning complete Urdu stories into right-to-left comics; vs baseline: spoken lines printed 0.49 → 1.00, dialogue F1 0.63 → 1.00, face consistency 0.54 → 0.67 (paired bootstrap).

Facial emotion recognition on imbalanced FER2013: CNN vs Vision Transformer (paper, pending publication)

- CNN (SE blocks) vs ViT-Base/16 on FER2013 with class-balanced focal loss; ViT avoids minority-class collapse (Disgust 64.9% vs 0%).

EDUCATION

MS Data Science, Quaid-e-Azam University, Islamabad — Feb 2025 – Present
Coursework: Machine Learning, Deep Learning, NLP, Big Data Analytics, Statistical Modeling, Data Visualization. Thesis: UrduComicLab.

CERTIFICATIONS

Machine Learning Specialization · Deep Learning Specialization · Natural Language Processing · Big Data Analytics